

METROPOLIS

There Is No Technology Fix for Car Bloat

A proposed rule could force automakers to downsize the largest of their vehicles. That's what's really needed to save pedestrians.

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Vishva Patel/Pexels

The number of pedestrian deaths in the United States is skyrocketing. In 2022 traffic crashes killed 7,805 people on foot—that's an 83 percent rise from 2009, and a 40-year high. The vast majority of those deaths involved a car colliding into a human.

In September, the National Highway Traffic Safety Administration took a significant step toward addressing the crisis of killer cars, proposing a new federal rule that would require—not ask—carmakers to ensure that the front ends of their vehicles do not create excessive risk of pedestrian head injuries. Should the proposal become law, hulking SUVs and pickups would face particular challenges passing NHTSA’s mandatory tests. Some cars would *have* to get a little bit smaller.

This wouldn’t solve car bloat; the head is just one part of the body that can become damaged by a vehicle. But it is a watershed moment for the NHTSA, the first time the agency has demanded that automakers adjust the physical shape of their vehicles to mitigate danger to those on foot. Currently, the NHTSA is soliciting comment, with the general public and automakers able to weigh in and help determine whether the rule ultimately goes on the books or gets watered down. The country stands at an important juncture, where it could finally start fixing the deadly problem of large cars—or allow them to continue their expansion.

Auto executives, for their part, do usually (though not always) acknowledge the role that their products have played in the American pedestrian safety crisis, but they favor a very different response. Rather than directly address vehicle size, they tout emergent technologies, like pedestrian automatic emergency braking, that can theoretically avoid crashes entirely, negating the risk that hulking vehicles present in a collision.

But there’s a problem: Those technologies don’t work consistently, and even in the future their benefits will be limited. The one reliable way to address the harms of gigantic SUVs and pickups is to stop building them.

I use the term car bloat to describe the ongoing expansion of vehicle models over the past 50 years. Although car bloat is a global trend, it is especially pronounced in the United States, where sedans and station wagons have been largely replaced by the SUVs and pickups that now account for about 4 in 5 new car purchases. At the same time, individual models have grown heftier. A 2024 Chevrolet Silverado pickup, for instance, is around 700 pounds heavier and 2 inches taller than the 1995 edition. According to federal data, the average new American car now weighs around 30 percent more than it did 40 years ago.

Car bloat creates numerous costs that are borne by society rather than the purchaser, or “negative externalities,” as economists call them. These include increased emissions, faster road wear, and reduced curbside parking capacity. But car bloat’s most obvious and urgent downside may be the danger it presents to anyone on the street who isn’t cocooned inside a gigantic vehicle.

Although occupants of big cars may be slightly safer in a crash, those in smaller ones are at much greater risk. A recent analysis by the Economist found that among the heaviest one percent of American cars, 12 people die inside smaller models for each person saved by the enormity of their vehicle. Pedestrians are still more exposed. A recent study by the Insurance Institute for Highway Safety found that vehicles with tall, flat front ends—common on SUVs and pickups—are more than 40 percent more likely to kill a pedestrian in the event of a crash than those with shorter, sloped ones. Worse, giant cars are more apt to hit a human in the first place because drivers sitting high off the ground have an obscured view of their surroundings. A 2022 IIHS study found that large vehicles' A-pillars (the structure between a windshield and side window) frequently conceal pedestrians at intersections, and TV news stations have run segments demonstrating that an SUV driver cannot see as many as nine toddlers sitting in a row in front of her.

The intuitive way to mitigate such pedestrian safety risks would be to shrink the size of the largest car models. But American automakers have shown little desire to do so because big SUVs and trucks are disproportionately profitable.

Instead, the companies have pointed to technologies that can supposedly negate car bloat's safety issues. Their executives often cite pedestrian automatic emergency braking, systems that use computer vision, lidar, or radar to identify a person ahead and halt the vehicle, preventing a crash. Earlier this year, the NHTSA finalized rules that will set minimum standards for PAEB on new cars, starting in 2029. When the NHTSA announced its new proposal to reduce the risk of pedestrian head injuries in a collision, the Alliance for Automotive Innovation, a trade group, cited PAEB in its response, implying that further regulations may be superfluous.

PAEB does have real value: Another 2022 study by IIHS found that existing PAEB systems could prevent about a quarter of pedestrian deaths. But that same study also found that automakers' PAEB systems did not work reliably while turning or at night, when around three-quarters of pedestrian deaths occur. A study published earlier this year by Missy Cummings, a robotics professor at George Mason University and a former NHTSA adviser, tested six vehicles' PAEB systems and concluded that the cars “were not consistent internally or with one another in pedestrian detection and response.”*

Cummings told me that those results did not surprise her. “PAEBs can detect pedestrians in some conditions, but because of their architecture, they inevitably cannot work 100 percent of the time,” she said. “Radar requires movement, so if it’s seeing something move, it can do a good job. Lidar requires dry weather, so if it has that, it does a great job.” She paused before passing judgment on computer vision, the centerpiece of Tesla’s automatic braking systems: “It never does a good job.”

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Automakers themselves seem to recognize the limitations of their pedestrian detection systems. [Chevrolet's PAEB webpage](#), for instance, presents a litany of caveats, warning that the technology may not detect children, people walking at night, or those who are part of a group.

To be fair, car companies are working to improve their PAEB systems in advance of the 2029 deadline that NHTSA established earlier this year. But even if those systems eventually work flawlessly, the laws of physics will prevent them from halting a fast-moving vehicle in time to avoid colliding with a pedestrian. NHTSA's 2029 PAEB standard will require cars to avoid collisions at 37 mph, but it sets no such expectation if the car is going faster. By the [agency's own estimates](#), around half of all pedestrian deaths occur on roads with a speed limit of 45 mph or higher. In other words, PAEB is far from a pedestrian-safety panacea.

The same goes for another technology that some, including [Democratic Connecticut Sen. Richard Blumenthal](#), have suggested could counteract the hazards of car bloat: [front and rear cameras](#) that give drivers a picture of what lies ahead of or behind them. Although such cameras can help drivers see areas that would otherwise be hidden due to their vehicle's height and hood design, they cannot provide unfettered vision. "Blind-spot cameras still have blind spots," said Cummings. "You still get that visual cone, and you're still going to get things that are not in view." Research [has also found](#) that people react faster to objects viewed directly rather than through a reflection or screen, suggesting that drivers may need more time to react when observing an emergency situation (like a small child) through a camera than they would if seeing it directly.

Historically, the U.S. has [treated street safety](#) as a problem that technology can fix without requiring changes to the shape and bulk of even the most egregiously oversized vehicles. That is why it is so significant that the NHTSA now wants to require automakers to ensure that the designs of their hoods do not pose excessive risk of pedestrian head injury. Further rules, including those pertaining to pedestrian torso impacts and cyclist crashes, could follow.

The agency is finally acknowledging that carmakers cannot simply innovate their way out of their responsibility to address the U.S. road safety crisis. Regulating car bloat directly is as appropriate as it is necessary. 

Correction, Oct. 2, 2024: This article originally misstated that Cummings tested three cars. She tested six.

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